

Case Study: COVID-19 Data Analysis

Overview:

The **COVID-19 Dashboard** provides an analytical view of the **pandemic's impact** by analyzing **key metrics** such as active cases, total deaths, recovered cases, and new infections worldwide. The dataset includes data from multiple WHO regions and highlights trends in infections, deaths, and recoveries.

Key Metrics Analyzed:

1. **Total WHO Regions:** 6 regions
 2. **Total New Cases:** 229K
 3. **Total Active Cases:** 6M
 4. **Highest Death Country:** USA
 5. **Death & Recovery Trends by Country**
 6. **1-Week % Increase in Cases by Region**
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Data Analysis & Insights:

1) How many WHO regions are present in the dataset?

☒ The dataset consists of **6 WHO regions**, including **Africa, Americas, Eastern Mediterranean, Europe, South-East Asia, and Western Pacific**.

2) What is the total number of new COVID-19 cases?

☒ The dataset records a total of **229K new cases**, indicating a significant spread of the virus in the observed period.

3) Which country has reported the highest number of deaths?

☒ The **United States (US)** has the **highest death toll**, highlighting its severe impact during the pandemic.

4) What is the total number of active cases?

☒ The dashboard reports **6 million active cases**, meaning individuals are still infected and undergoing treatment.

5) What is the trend of deaths and recoveries by country?

☒ The **bar chart analysis** shows that countries like the **US, Brazil, India, and Italy** have high death counts, while some countries show a higher recovery rate than deaths.

6) Which region has the highest 1-week % increase in cases?

☒ The **Europe region** has the **highest 1-week increase in cases** at 56%, followed by **Africa (48%)** and **the Americas (35%)**.

Final Conclusion & Recommendations:

✦ **High-risk countries like the US should focus on stronger public health policies** to control rising deaths.

✦ **WHO regions with high % case increases (Europe, Africa) should enhance vaccination & testing programs.**

✦ **Further analysis should be done on the impact of lockdowns and healthcare system preparedness.**